

Rajdeep Chakraborty

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 LinkedIn |  Github |  Website

RESEARCH EXPERIENCE

- **Indian Institute of Science** Jul 2024 - Apr 2025
Research Assistant Bengaluru, India
 - Investigated the role of the Superior Colliculus (SC) in mediating head orientation towards noxious stimuli using transgenic mice and chemogenetic approaches.
 - Wrote Python scripts to automate data collection, performed statistical analyses, and trained machine learning models to study and classify mice behavioral patterns.
 - Led a 3-member team to design and automate a behavioral assay framework, enabling high-throughput investigation of stress-driven decision-making in mice.
 - Optimized and standardized experimental protocols, improving reproducibility and reliability of behavioral assays.

EDUCATION

- **Maulana Abul Kalam Azad University of Technology** 2018 - 2022
Bachelor of Technology in Mechanical Engineering Kolkata, India

PROJECTS

- **Project 1: Automated Mice Assay** Jan 2025 - Apr 2025
Tools: [Arduino, Python, KiCad, PLX-DAQ, Excel]  
 - Developed an automated behavioral assay using Arduino to study mouse decision-making behavior under different stress-reward scenarios.
 - Applied circuit design principles to automate decision gates, stimulus delivery and reward-dispensing mechanisms to achieve precise trial control.
 - Implemented a Python-based data pipeline to record, back up, and analyze behavioral data, including session-wise performance tracking and visualization.
- **Project 2: Predicting Personality Traits From Cognitive Task Performance** Jul 2023 
Tools: [Python, Data Analysis, Machine Learning]
 - Created a decoding model that integrated behavioral scores (Inverse Efficiency Score) and task-based fMRI contrast maps, ensuring multi-modal personality prediction.
 - Applied concepts and methods taught during the summer school program, like Machine Learning, Model Evaluation.
 - Worked with a group of six members from diverse backgrounds and collaboratively presented the findings to an international audience of researchers and peers.

SKILLS

- **Programming Languages:** Python, MATLAB, R
- **Statistical & Data Analysis Libraries:** pandas, numpy, seaborn, matplotlib, statsmodels, scipy
- **Machine Learning Libraries:** scikit-learn, Tensorflow
- **Statistical Methods:** Inferential Statistics, Descriptive Statistics, Statistical Modeling, Hypothesis Testing
- **Research Tools & Platforms:** DeepLabCut, ImageJ, Git, GraphPad Prism
- **Technical Skills:** Microcontrollers, Scripting, Experimental Design, Version Control
- **Research Skills:** Data & Statistical Analysis, Scientific Presentation

PROFESSIONAL EXPERIENCE

- **Packt Publishing**

Oct 2023 - Jun 2024

Technical Editor

Remote

- Executed code snippets and applied problem-solving skills along with research to ensure technical proficiency.
- Collaborated with authors, editors and reviewers to maintain the timely delivery of accurate content.
- Managed book repositories and addressed customer-reported issues for accuracy and clarity.

- **Focus Edumatics**

Aug 2022 - May 2023

Online Mathematics Tutor

Remote

- Delivered personalised tutoring sessions for grade 5-12 students with an average rating of **4.9/5**.
- Crafting lesson plans through in-depth exploration of mathematical concepts and utilising the Learning Management System.
- Strengthening teaching methodologies and abilities by assisting students in tackling homework assignments successfully.

VOLUNTEER EXPERIENCE

- **Student Outreach Volunteer**

Sept 2024 - Dec 2024

Indian Institute of Science

- Volunteered for Department Outreach Program and delivered talks on **Introduction to Neuroscience** to student audience.
- Collaborated with peers to organise department-wide outreach event.
- **Skills:** Public Speaking, Science Communication

CERTIFICATIONS

- **MITX:** Quantitative Biology Workshop

Dec 2024

- **Neuromatch Academy:** Computational Neuroscience

Jul 2023

- **HarvardX:** Fundamentals of Neuroscience

May 2023

- **MITX:** Computational Thinking using Python

Dec 2022

- **HarvardX:** Using Python for Research

Sept 2022